



Tour de Maths 2023

Leg 5

Hosted by:



BEAULIEU COLLEGE

11 May 2023

Section A (5 mark questions)

Question 1

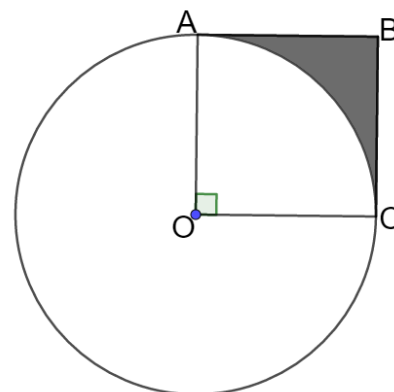
Which of these numbers is closest to 1?

$$\frac{8}{9} \text{ or } \frac{9}{8} \text{ or } \frac{10}{11} \text{ or } \frac{11}{10}$$

Question 2

In the diagram alongside. A and C lie on the circumference of the circle with centre O. OABC is a square with area 16 cm^2 .

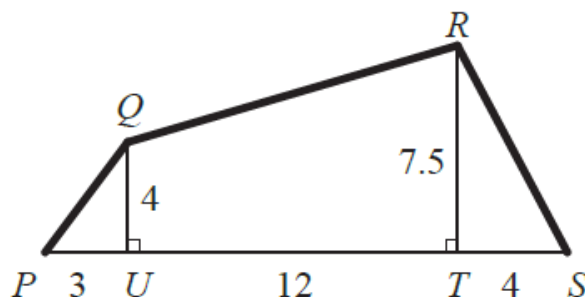
What is the area of the shaded region correct to two decimal places?



Question 3

A large tent is being set up for a fair. Two poles, QU and RT, are placed perpendicular to the ground and 12 m apart. Pole QU is 4 m in length and pole RT is 7.5 m in length. A tarp is placed over the poles and secured to the ground at P, 3 m from the base of pole QU, and S, 4 m from the base of pole RT.

Determine $PQ + QR + RS$, the length of the tarp.



Question 4

The rectangular floor plan of the first level of a house is shown in the diagram alongside. Both the laundry room and the dining room are squares with areas of 4 m^2 and 25 m^2 , respectively. The living room is rectangular with an area of 30 m^2 . Determine the area of the kitchen.



Question 5

If $2^2 \times 3^3 \times 4^4 \times 5^{11}$ is multiplied out, what is the sum of the digits of the result?

Question 6

In a traffic study done on a busy highway, it was found that 40% of all cars contained two or more people. Of those cars containing only one person, 25% of these people were NOT wearing sunglasses.

Determine the percentage of cars that contained exactly one person and that that person was wearing sunglasses.



Question 7

Four students share a R1000 prize for a project. They decide to split the money based on the amount of work each team

member contributed to the project. Alison and Bob receive $\frac{1}{8}$

and $\frac{1}{5}$ of the total respectively. Carl receives the average of what Alison and Bob receive.

Dana receives the remainder. Determine the fraction of the prize money that Dana receives.



Question 8

The dimensions of a rectangular prism (box) are x cm by x cm by h cm. If the surface area of the prism is $14x^2$ cm², express h in terms of x .

Question 9

The mean (average) mark for 10 pupils in class A for a test is 57%. Another class, B, has a mean of 63%. The mean of the two classes combined is 61%. How many pupils are there in class B?

Question 10

Below there is a list of numbers with their codes. Determine the three-digit number w .

Number	Code
589	521
724	386
1346	9764
w	485

Section B (10 Mark questions)

Question 11

What is the last digit of $2^{2018} - 2$?

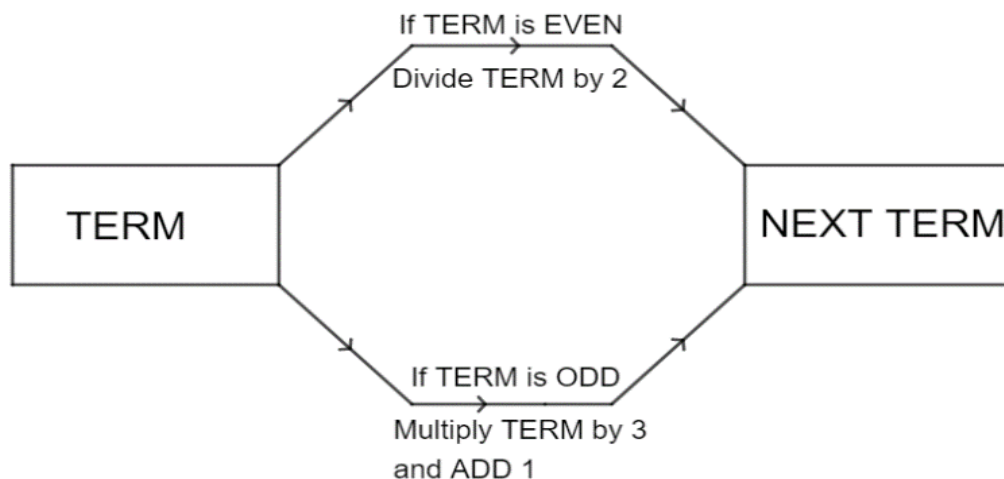
Question 12

In a sequence of numbers, each number in the sequence is called a term. In the sequence: 1; 3; 5; 7; 9, the first term is 1, the second term is 3, the third term is 5, the fourth term is 7 and the fifth term is 9.

In another sequence, the first term is 26. If a term in the sequence is even, then the next term will be half the value of that term. If a term in the sequence is odd, then the next term in the sequence is one more than three times that term. The rules are illustrated in the diagram below.

By following these rules, the first three terms of this sequence are 26; 13 and 40.

What is the value of the 35th term of the sequence?



Question 13

Salim wants to download a certain number of his favourite tunes using some money he has saved doing odd jobs. At Downloads For U, they charge R1,00 per download. He would need to save R17,00 more than he already has in order to download all he wants. At a competing site, Music Bytes, they only charge R0,80 per download. He would have R5,00 left over from his savings if he downloads all he wants.

How many downloads does Salim want?

Question 14

The decimal expansion of $\frac{1}{7}$ is the recurring decimal $0,142857$. What digit occurs in the 2018th place after the decimal coma?

Question 15

Ms Payn planted five rose plants in a row along one side of his property. She then planted one tulip plant in each of the spaces in the row between the roses. Next, she planted one daffodil plant in each of the spaces between the plants already in the row. She then repeated this procedure with daisies, then marigolds, and finally with lilies. Determine the total number of plants in the row.



Section C (20 mark questions)

Question 16

Find the value of : $\sqrt{1^3 + 2^3 + 3^3 + 4^3 + 5^3 + \dots + 100^3}$

Question 17

On your 13th birthday, you received three different time-travel pedometers. You want to use your pedometers to travel back to your 8th birthday. You may use your pedometers as often as you wish but only one at a time.

- Each time you use Pedometer A, take exactly 7 steps forward. This will result in you going back 4 months in time.
- Each time you use Pedometer B, take exactly 5 steps backward. This will result in you going back 7 months in time.
- Each time you use Pedometer C, take exactly 2 steps forward. This will result in you going back 3 months in time.

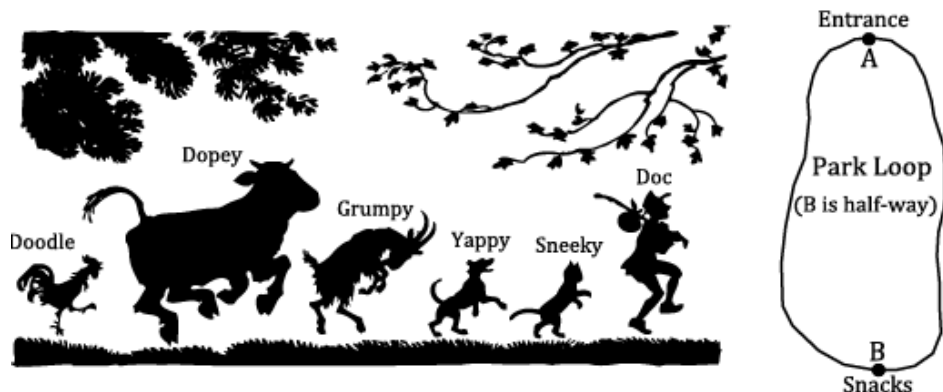
In traveling back to your 8th birthday, you made a total of 25 backward steps and had a total of 12 pedometer uses.

How many forward steps did you take while using your pedometers?

Question 18

Doc takes his menagerie on a long walk, parading along a 2 km loop around the park. Doc walks at a leisurely 4 km/h^{-1} , as do Sneeky, Yappy and Doodle. However, Grumpy and Dopey walk, in the same direction, at 6 km/h^{-1} .

If they walk for $2\frac{1}{2}$ hours, how many times will Grumpy and Dopey lap Doc and his friends Sneeky, Yappy and Doodle?



Question 19

A large jug of lemonade contains a mixture of lemon juice concentrate and water. When 1 litre of water is added to the jug, the ratio, by volume, of juice concentrate to water is 1:2. When 1 litre of juice concentrate is added to the new mixture, the ratio becomes 2:3. Determine the original ratio, by volume, of lemon juice concentrate to water in the jug.



Question 20

A hardware store sells single digits to be used for house numbers. There are five 5s, four 4s, three 3s and two 2s available.

From this selection of digits, a customer is able to purchase her three-digit house number. Determine the number of three-digit house numbers this customer could form.

